

LW4 Tutorial: Integer Programming

Task 1 University Committee Selection Problem:

A university is forming a new Student–Faculty Relations Committee. There are 10 candidates available, each of whom may belong to multiple categories (female, male, student, administrator, faculty). The goal is to form the smallest possible committee that satisfies the university’s representation requirements.

The committee must include at least one member from each of the following categories: Female, Male, Student, Administrator and Faculty. The category membership of each candidate is listed in the table below.

Table 1. Membership Category Table

Category	Individuals
Females	1,2,3,4,5
Males	6,7,8,9,10
Students	1,2,3,10
Administrators	5,6
Faculty	4,7,8,9

In addition, the university has imposed the following special constraints:

- Candidates 5 and 6 form a pair which means they are both selected or none.
- Candidate 6 may only be selected if candidate 8 is also selected.
- From the set {1, 2, 3, 4, 5}, at most one candidate may be selected.

Task:

- Step 1: Based on the above Membership Category Table, convert the membership information into a structured, parameter-based format.
- Step 2: Formulate the problem by clearly define the decision variables, objective function and constraints.
- Step 2: Implement the model in Excel and solve it by the Solver. How do you interpret the solution?

Task 2 Emergency vehicles location problem

Emergency coverage involves deciding the location of emergency vehicles around a city with various districts as potential locations. The city is divided into 9 districts, and 7 sites have been identified as potential locations for the emergency vehicles. A vehicle located in a given site can reach some of the districts within the required time limit as indicated in the table. The problem is to provide emergency vehicle coverage to all the 9 districts within the required time while using the minimum number of emergency vehicles.

The table of district–site coverage is as follows:

	S1	S2	S3	S4	S5	S6	S7
D1	0	1	0	1	0	0	1
D2	1	0	0	0	0	1	1
D3	0	1	0	0	0	1	1
D4	0	1	1	0	1	1	0
D5	1	0	1	0	1	0	0
D6	1	0	0	1	0	1	0
D7	1	0	0	0	0	0	1
D8	0	0	1	1	1	0	0
D9	1	0	0	0	1	0	0

Task 3 Swimmer team build problem

A group of swimmers must be assigned to a 200-meter medley relay team, consisting of four 50-meter segments. Most swimmers are very fast in more than one stroke, so it is not clear which swimmer should be assigned to each of the four strokes. The five best swimmers and their times for each stroke are provided in the following table. The problem is to assign four swimmers to the four strokes in such a way that the total time is minimised.

Stroke	SW1	SW2	SW3	SW4	SW5
Backstroke	37.7	32.9	33.8	37.0	35.4
Breaststroke	43.4	33.1	42.2	34.7	41.8
Butterfly	33.3	28.5	38.9	30.4	33.6
Freestyle	29.2	26.4	29.6	28.5	31.1

Task 4: Formulate the algebraic model for the various optimisation problems from the lecture notes: Project Budgeting (slide 37), Software Components (slide 38), and Capital Investments (slide 39).